

Due on tuesday after Study Break, April 3rd, in class

This problem set is double the usual size. Please start work early.

If marked, half of the problem set will be marked.

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1. Consider a set of inertial frames which are aligned at the commonly defined zero time, but might be moving in different relative directions.
 - (a) [**1+3 pts**] Write down the Lorentz boost matrices for relative velocity v_1 between frames in the x -direction, and for relative velocity v_2 in the y -direction.
 - (b) [**7 pts**] Find out by matrix multiplication the result of successive application of these two transformations. Consider both possible orderings: applying the x -direction boost first and then the y -direction boost, and vice versa.
This will look cumbersome, so I suggest that you use $c = 1$ units and the notation $\gamma_1 = \gamma(v_1)$, $\gamma_2 = \gamma(v_2)$. Another hint: do you really need to multiply 4×4 matrices?
 - (c) [**4 pts**] Do the two Lorentz boost matrices commute? Compare with Lorentz boosts in the same direction.
 - (d) [**1+2 pts**] Consider any of the matrices resulting from successive application of the two boosts. Is the matrix symmetric?
Remembering from Problem Set 5 that Lorentz boost transformations in arbitrary directions are symmetric matrices, interpret this observation.
2. The kinetic energy of motion of a particle is the relativistic total energy minus the rest energy.
 - (a) [**5 pts**] A particle has rest mass M and speed v . If $v \ll c$, then show that the kinetic energy due to motion is approximated by the well-known non-relativistic expression for the kinetic energy.
 - (b) [**5 pts**] Derive the leading correction to the non-relativistic expression.
 - (c) [**5 pts**] What is a particle's speed if its kinetic energy T is equal to its rest energy? What is a particle's speed if its kinetic energy is n times its rest energy?
 - (d) [**4 pts**] Express the momentum p of the particle as a function of the mass M and the kinetic energy of motion, which we will call T . Your expression should contain M and T , not v or γ_v .
Hint: use the relation between E and p that does not involve v or γ_v .

3. Some groups. This problem requires some searching and reading.

Look up the terms $SO(2)$, $SO(3)$, $SO(3,1)$, $SU(2)$. Each of these groups can be thought of as a set of matrices.

- (a) [**2 pts.**] What does ‘ S ’ in these names stand for?
Which matrices are included in the group $O(3)$, but not in the group $SO(3)$?
- (b) [**3 pts.**] Which physical transformations are represented by the groups $SO(2)$ and $SO(3)$? Are these groups abelian or non-abelian?
- (c) [**1 pt.**] What is the physical interpretation of matrices which are included in the group $O(3)$, but not in the group $SO(3)$?

4. Spacetime diagrams (plots of ct versus x).

- (a) [**4 pts.**] On a spacetime diagram, draw the worldline of a photon starting at the origin, the worldline of an object with speed $c/2$ starting at the same time at the origin, and the worldline of an object with speed $4c$ starting at the same time at the origin. Which worldline is unphysical?
- (b) [**3 pts.**] On a spacetime diagram, draw the worldline of a photon starting at the origin, and the worldline of an object with velocity $c/2$ starting at the origin four seconds later.

5. Frame Σ' moves at speed v with respect to frame Σ , in the common x, x' directions. The two frames are aligned at time $t = t' = 0$.

- (a) [**2 pts.**] Write down the Galilean transformation as a matrix transforming the Σ -frame coordinates (ct, x) to the Σ' coordinates (ct', x') . You can ignore the transverse directions and work with 2×2 matrices.
- (b) [**4 pts.**] Write down the Lorentz transformation matrix. Take the limit $c \rightarrow \infty$, and show that this fails to reproduce the Galilean transformation matrix.
- (c) [**6 pts.**] The reason for the failure above is that in the $c \rightarrow \infty$ limit, the variables ct and ct' lose meaning.

So, write down the Galilean transformation as a matrix transforming the Σ -frame coordinates (t, x) to the Σ' coordinates (t', x') . i.e., the first coordinate does not have the c factor. Write down the Lorentz transformation the same way, as a matrix transforming (t, x) to (t', x') .

Now show that the Lorentz transformation matrix reduces to the Galilean transformation matrix in the limit $c \rightarrow \infty$.

- (d) [5 pts.] Considering two successive Galilean transformations in the same x direction, with velocities v_1 and v_2 , show by matrix multiplication that their combination is also a Galilean transformation, with velocity $v_1 + v_2$.

6. Relative velocities, longitudinal & transverse.

- (a) [4 pts] A rocket traveling at speed $\frac{1}{2}c$ with respect to frame Σ shoots forward bullets at speed $\frac{3}{4}c$ relative to the rocket. What is the speed of the bullets relative to Σ ?
- (b) [5 pts] Now imagine that the bullets are fired *perpendicular* to the direction of rocket motion, as perceived from the rocket. What is the speed of the bullets relative to Σ ?
- (c) [6+3 pts] An X-ray beam is sent out from the rocket, perpendicular to the direction of rocket motion. Relative to Σ , what are the components of the X-ray beam velocity parallel and perpendicular to the direction of rocket motion? Calculate from your velocity components the speed of the X-ray beam relative to Σ . Explain why the speed relative to Σ could have been expected.

7. This problem continues from question 3.

- (a) [1 pt.] How many matrices does $SO(2)$ contain?
- (b) [6 pts.] The group $SO(3, 1)$ is the proper Lorentz group. What is the size of matrices belonging to this group? These matrices are not orthogonal, but must instead satisfy a similar equation. Write down the equation that defines membership in the Lorentz group. Under what condition is a member of this group called *orthochronous*?
- (c) [2 pts.] What is the size of matrices belonging to the group $SO(1, 1)$? Which subset of Lorentz transformations forms this group?

8. Some more spacetime diagrams.

- (a) [3 pts] On a spacetime diagram, depict the situation that a light source at $x = 0$ is turned on for a few seconds, and then turned off again. (Show the worldlines of photons.)
- (b) [3 pts] A railgun located at the origin shoots projectiles at time intervals of T , alternately in the positive- x and negative- x directions. The projectiles in the positive x direction have speed $c/10$ and those in the negative x direction have speed $c/2$. Draw the situation on a spacetime diagram, showing the worldlines of a number of successive projectiles.